

virtual reality (OSVR). We have seen that it encourages experimentation of headsets development as well. Now, if you look at the other side of the coin, on the device layer you will see OpenXR standard being badly adopted by the manufacturers which allows applications and engines to target with a single application programming interface (API)," says Vinay G. Rajagopal, Red Hat

He continues, "We have already started seeing a proliferation of 3D environments that have been created by the users themselves. That is where the whole blockchain has become an imperative to ensure the details surrounding a transaction and the digital assets or the avatars. It has become important for us to put these assets on multiple ledgers while making sure that you understand who has built, sold, or purchased a particular digital asset. It becomes more important from a legal aspect too."

"Encouraging the development of a decentralised, distributed, interoperable virtual world built on the foundation of open standards also creates multiple opportunities for us as well," adds Vinay.

"Open source is the ground zero for the development of technologies like the metaverse. All of the biggest projects in the last five to ten years that have been able to garner the most attraction are the ones that are open sourced. Artificial intelligence (AI), blockchain, and containerisation are to name a few. Open source comes with a great number of benefits. It allows you to collaborate with the smartest minds from every corner of the world, your work gets constantly reviewed and used while having a constant feedback loop," says Gandhali Samant, GitHub, India.

"A good aspect of working with open source is that there will always be components ready to be used, which saves a lot of time to start from scratch. As the metaverse itself continues to gain attraction, more tools

and platforms will appear to assist the developers," adds Gandhali.

"Even though open source comes with a lot of benefits, such as quick development and huge communities, it is important to note these communities own the software itself. In 2014, a bug was detected in Unison. Its source code had been open for so long that it made it hard to estimate the time hackers had been exploiting it," says Dr Sumit Kalra, IIT Jodhpur. "Today, whatever we do with the metaverse is going to have a tremendous impact on the future of the metaverse development. If we are going to continue the development with open source, then it is crucial for us to also adopt open standards."

Syed Mohammed Khasim from Texas Instruments notes, "In the last couple of decades I have seen the evolution of open source starting from Linux. It all essentially started for shared software development."

"The cost of developing a software turns out to be a big challenge as it requires a unique skill set. You need to have a proper architecture to build a software, a design and the necessary components which will allow you to get started with the coding aspect of it. You will then have to validate and commercialise everything before it goes into the hands of developers and ultimately the consumers themselves. It all boils down to a sufficient investment. But there is a limit to which everyone can invest in a software, and often it is only for a particular period of time," he adds.

"Open source is no free lunch as somebody is indeed paying for it. When we use the term open source, we mean that we are enabling a shared software development model where we want to collaborate with similar entities that want to invest in the development of software," says Syed.

He adds, "Hardware is also one of the fundamental aspects of

building software as without proper hardware and place software cannot be developed. We need lowcost hardware that can be afforded by everybody. This is why the Intel ecosystem picked up because everyone uses a personal computer (PC) and these systems do not require a huge investment. Over time, embedded systems have also evolved and reduced a lot of the burden of software development. With the induction of Android all the necessary components of open source development have been stitched together."

"Today, things are different as they have evolved tremendously. We are seeing a multi-core architecture and multi-core programming even in a simple smartwatch today. It requires a great amount of skill to do it all. You are not only going to work for free while working on open source software but your contribution to open source will also provide you with the relevant validation and introduce you to new features. Hence, companies always try to build an ecosystem for open source development as it tends to be more robust and reliable," states Syed.

He further adds, "Now is the time to define the hardware ecosystem that we are going to enter into the kind of software skill-sets that are required so it can be assessed as to where we need to build the community around. We also need to find whether it can scale and sustain on its own without a huge sum of money being pumped from any big organisations."

Biju K. Nair, LegaliTech, mentions, "Open source as a whole has various aspects to it. There is also a very important legal aspect to it from an intellectual property (IP) perspective. Considering IP is different in metaverse than how it is considered in the traditional ecosystem, it also creates opportunities to maintain and safeguard the IP within the metaverse. In the last two years, the whole globe has woken up to